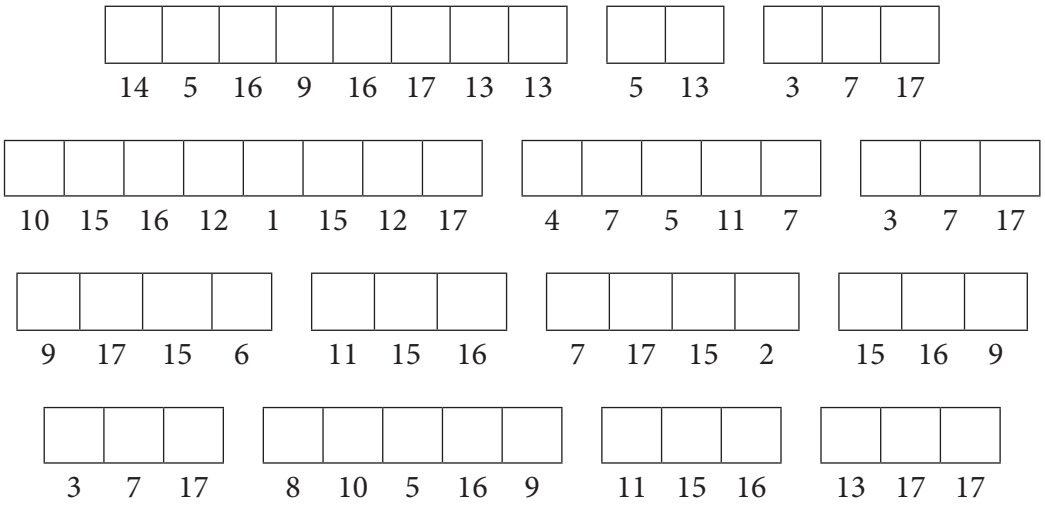
Name: _____

WORKSHEET

Quadratic formula



Mark Twain

Rule

Given $ax^2 + bx + c = 0$, we can solve for x using the quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Note: $b^2 - 4ac$ cannot be negative.

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Questions

Solve the following quadratic equations for x.

1 $x^2 + 7x + 12 = 0$

2 $2x^2 + 9x - 5 = 0$

3 $x^2 + 3x - 10 = 0$

4 $3x^2 - 5x + 2 = 0$

5 $10x^2 - 26x + 12 = 0$

6 $x^2 + 4x - 21 = 0$

7 $6x^2 - 13x - 5 = 0$

8 $2x^2 - 29x + 77 = 0$

9 $x^2 - 5x - 6 = 0$

10 $3x^2 - 20x - 32 = 0$

11 $3x^2 - 42x + 135 = 0$

12 $x^2 - 3x = 4$

13 $2x^2 - 3 = -5x$

14 $5x^2 = 3x + 2$

15 $12x = 4x^2 + 5$

16 $x^2 + 7x - 5 = 2x + 9$

17 $2x^2 + 11x = 20 - x^2$

Solutions

A $x = 2\frac{1}{2}, \frac{1}{2}$

B $x = 11, 3\frac{1}{2}$

C $x = 9, 5$

D $x = 6, -1$

E $x = 1\frac{1}{3}, -5$

F $x = 3, -7$

G $x = 4, -1$

H $x = 2\frac{1}{2}, -\frac{1}{3}$

I $x = 2, \frac{3}{5}$

K $x = 1, -\frac{2}{5}$

L $x = 8, -1\frac{1}{3}$

N $x = 2, -7$

R $x = \frac{1}{2}, -5$

S $x = \frac{1}{2}, -3$

T $x = 2, -5$

U $x = -3, -4$

W $x = 1, \frac{2}{3}$

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